

Problem 1:

A space X is *locally metrizable* if each point in X has a neighborhood which is metrizable in the subspace topology. Prove that if X is a locally metrizable compact Hausdorff space, then it is metrizable. *Hint:* you should use the Urysohn metrization theorem; for a further hint, see Exercise 7 in §34 of Munkres.

Proof.

Assume X is locally metrizable compact Hausdorff space. Thus, by homework X is normal since it is compact and Hausdorff. Therefore, X is also regular.

Let $x \in X$. Let U_x be a neighborhood of x s.t. it is metrizable. Then define $B_x = \{U_x \cap B(x, 1/n) \mid n \in \mathbb{N}\} \cup \{U_x \cap B(x, n) \mid n \in \mathbb{N}\}$.

Consider $\{U_x\}_{x \in X}$. This is an open cover of X . Thus, there is a finite subcover, $U_{x_1}, \dots, U_{x_n}$ since X is compact.

Since U_x is open in X and $B(x, \epsilon)$ is open in U_x . This implies that $B(x, \epsilon) = U_x \cap O$ for O open in X . Thus, $U_x \cap B(x, \epsilon) = U_x \cap O$ is open in X .

Define β as all finite intersections of $B = B_{x_1} \cup \dots \cup B_{x_n}$

Basis:

Let $x \in X$. Then $x \in U_{x_i}$ for some i . Thus, let d be the metric for U_{x_i} . Thus, $d(x, x_i) = a$ for some $a \in \mathbb{R}$. Thus, fix $N \in \mathbb{N}$ s.t. $N > a$. Thus, $x \in B(x_i, N) \cap U_{x_i} \in \beta$.

Let $x \in B_1 \cap B_2$ s.t. $B_1 \cap B_2 \in \beta$. Thus, $B_1 \cap B_2 \in \beta$. Since B_1 is a finite intersection of things in B and same with B_2 , so their intersection is also a finite intersection.

Thus, $x \in B_1 \cap B_2 \subset B_1 \cap B_2$.

Thus, β is a countable basis for X .

Thus, by the Urysohn metrization theorem we know that X is metrizable. ■

Problem 2:

Let X be a topological space and let Y be a metric space. A function $f : X \rightarrow Y$ is *bounded* if there exists some $R > 0$ such that, for any $x_1, x_2 \in X$, we have $d(f(x_1), f(x_2)) \leq R$. The set of all bounded functions $f : X \rightarrow Y$ is denoted $\mathcal{B}(X, Y)$.

(a)

Is $\mathcal{B}(X, Y)$ necessarily closed in the topology of pointwise convergence on Y^X ?

Proof.

No consider $\mathbb{R}^{\mathbb{R}}$

$$f_n(x) = \begin{cases} x & -n \leq x \leq n, \\ -n & -n < x, \\ n & n > x. \end{cases}$$

$f_n \in \mathcal{B}(\mathbb{R}, \mathbb{R})$ but $f_n \rightarrow f(x) = x$ since fix $\epsilon > 0$, and $a \in \mathbb{R}$.

Fix $N \in \mathbb{N}$ big enough s.t. $|a| < N$. Thus, $d(a, f(a)) = 0 < \epsilon$. Thus, $f_x(x) \rightarrow f(x)$ for all $x \in \mathbb{R}$. Thus, $f_x \rightarrow f$, but f is not bounded, so $\mathcal{B}(X, Y)$ is not closed since if it was closed it would contain all of its limit points. ■

(b)

Is $\mathcal{B}(X, Y)$ necessarily closed in the topology of uniform convergence on Y^X ?

Proof.

Let (f_n) be a convergent sequence of functions in $\mathcal{B}(X, Y)$ s.t. $(f_n) \rightarrow f$. Fix $\epsilon > 0$. Thus, there is an $N \in \mathbb{N}$ s.t. $d(f_n(x), f(x)) < \epsilon$ for all $n \geq N$ and for all $x \in X$.

Thus, consider $d(f(x), f(y))$ s.t. $x, y \in X$. By the triangle inequality

$$\begin{aligned} d(f(x), f(y)) &\leq d(f(x), f_N(x)) + d(f(y), f_N(y)) \\ &\leq d(f(x), f_n(x)) + d(f(y), f_N(y)) + d(f_N(x), f_N(y)) \\ &\leq 2\epsilon + R_N \end{aligned}$$

for some $R_N \in \mathbb{R}$ since f_N is bounded.

Thus, f is bounded. Since $\mathcal{B}(X, Y)$ contains all of its limit points this implies that it is closed. ■

Problem 3:

Let X and Y be metric spaces; recall that $\mathcal{C}(X, Y)$ denotes the set of continuous functions $X \rightarrow Y$. There is a map $e : X \times \mathcal{C}(X, Y) \rightarrow Y$ defined by the rule:

$$e(x, f) = f(x).$$

This map is called the *evaluation map*. Prove that, if $\mathcal{C}(X, Y)$ has the topology of uniform convergence, then e is a continuous map.

Proof.

Let $U \subset Y$ be open. Let $(x_0, f_0) \in e^{-1}(U)$.

Thus, $f_0(x_0) \in U$. Therefore, there exists $\epsilon > 0$ s.t. $B(f_0(x_0), \epsilon) \subset U$.

Since f_0 is continuous, there exists an open neighborhood $V \subset X$ of x_0 s.t.

$f(V) \subset B(f_0(x_0), \epsilon/2)$.

Consider the open ball $B(f_0, \epsilon/2) \subset \mathcal{C}(X, Y)$ this implies that $d_Y(f(x), f_0(x)) \leq \epsilon/2$ for all $x \in V$.

Thus, for $(x, f) \in V \times W$

$$d_Y(f(x), f_0(x)) \leq d_Y(f(x), f_0(x)) + d_Y(f_0(x), f_0(x_0)) \leq \epsilon$$

Thus, $f(x) \in B(f_0(x_0), \epsilon) \subset U$.

Thus, $(x_0, f_0) \in V \times W \subset e^{-1}(U)$. Therefore, $e^{-1}(U)$ is open. ■

Problem 4:

Let X be a space, and let F be the set of all functions $X \rightarrow \mathbb{R}$. Suppose that the uniform topology and the point-open topology on F agree. What must be true about X ?

Proof.

Assume for contradiction that X is infinite. Thus, let $\{x_n\}$ be a countably infinite subset of X .

Define $f_n(x) = 1$ if $x = x_n$ or 0 otherwise.

Fix $\epsilon > 0$ and $x \in X$.

If $x \notin \{x_n\}$. Then $d(f(x), 0) = 0 < \epsilon$ for all $n \in \mathbb{N}$.

If $x \in \{x_n\}$ then $x = x_i$. Fix $N \in \mathbb{N}$ s.t. $N > i$. Thus, $d(f_n(x), 0) = 0 < \epsilon$ for all $n \geq N$.

Thus, $(f_n) \rightarrow 0$.

However, $\sup_{x \in X} d(f_n(x) - 0) = 1$ for all $n \in \mathbb{N}$. Thus, f_n does not converge uniformly to 0.

Thus, this is a contradiction thus X is finite. ■

Problem 5:

Consider the sequence of functions $f_n : (-1, 1) \rightarrow \mathbb{R}$ defined by

$$f_n(x) = \sum_{k=1}^n kx^k.$$

(a)

Prove that f_n converges pointwise, but *not* uniformly.

Proof.

Consider the power series

$$\sum_{k=1}^{\infty} x^k = \frac{1}{1-x} \quad \text{for } |x| < 1$$

Thus, taking the derivative and multiplying by x we get.

$$\sum_{k=1}^{\infty} kx^k = \frac{x}{(1-x)^2} \quad \text{for } |x| < 1$$

Thus, let $f(x) = \frac{x}{(1-x)^2}$.

Fix $\epsilon > 0$ and $x \in (-1, 1)$

Thus,

$$\begin{aligned} |f(x) - f_n(x)| &= \left| \sum_{k=n+1}^{\infty} kx^k \right| \leq \sum_{k=n+1}^{\infty} k|x|^k = \sum_{j=0}^{\infty} (n+1+j)|x|^{n+1+j} \\ &= |x|^{n+1} \left((n+1) \sum_{j=0}^{\infty} |x|^j + \sum_{j=0}^{\infty} j|x|^j \right) = |x|^{n+1} \left(\frac{n+1}{1-|x|} + \frac{|x|}{(1-|x|)^2} \right) \end{aligned}$$

Since $(|x|^{n+1}(n+1)) \rightarrow 0$ and $(C|x|^{n+1}) \rightarrow 0$ for $C = |x|/(1-|x|)^2$. This is because $|x| < 1$.

Thus, there is an $N_1, N_2 \in \mathbb{N}$ s.t. $|x|^{n+1}(n+1) < \epsilon/2$ and $C|x|^{n+1} < \epsilon/2$.

For the last part use $N = \max\{N_1, N_2\}$

Thus,

$$|x|^{n+1} \left(\frac{n+1}{1-|x|} + \frac{|x|}{(1-|x|)^2} \right) < \epsilon$$

Therefore, $f_n(x)$ converges pointwise to f on $(-1, 1)$

Suppose f_n converges uniformly to f on $(-1, 1)$. Then for $\epsilon = 1$ there exists an $N \in \mathbb{N}$

s.t. for all $n \geq N$ and all $x \in (-1, 1)$

$$|f_n(x) - f(x)| < 1$$

Shown above we know that $(n+1)x^{n+1} \leq |f_n(x) - f(x)| = R(x) < 1$.

Consider $x_n = 1 - (1/n)$

Thus, $(n+1)(1 - 1/n)^{n+1} \leq R(x_n)$.

Thus, as $n \rightarrow \infty$ shown in analysis $(1 - 1/n)^{n+1} \rightarrow e^{-1}$.

Therefore, there exists an $N_1 \in \mathbb{N}$ s.t. for all $n \geq N_1$

$$-1/2e < (1 - 1/n)^{n+1} - 1/e < 1/2e \implies 1/2e < (1 - 1/n)^{n+1}.$$

Thus, for such $n \geq N_1$, $R(x_n) \geq (n+1)/(2e)$.

Also for $(n+1)/(2e)$ consider $N_2 = 3$ s.t. for all $n \geq N_2$ $(n+1)/2e > 1$.

Thus, for $\max N, N_1, N_2$ we have $R(x_n) \geq (n+1)/2e > 1$. This is a contradiction thus, f_n does not uniformly converge. ■

(b)

Prove that f_n converges in the topology of compact convergence, so its limit is a continuous function.

Proof.

Let $K \subset (-1, 1)$ since $(-1, 1)$ is a subspace of $\mathbb{R}$. This implies that K is closed and bounded. Thus, there exists an $r \in (0, 1)$ s.t. $K \subset [-r, r]$.

Let $M_k = kr^k$.

Define a new sequence $g_n(x) = nx^n$.

Thus, for $x \in K$ we have $|g_n(x)| \leq n|x|^n \leq M_n$.

Since $\sum_{k=1}^{\infty} M_k$ converges by standard power series rules which is shown above since $|r| < 1$. We get that by the Weierstrass M-test that $\sum_{k=1}^{\infty} g_n(x)$ converges uniformly on $[-r, r]$.

Thus, this implies that $f_n(x)$ converges uniformly to on $K \subset [-r, r]$.

Also, since $(-1, 1)$ is a subspace of $\mathbb{R}$ it is metrizable thus first countable. Thus, it is compactly generated. Also since $\mathbb{R}$ is metrizable this implies that $\mathcal{C}((-1, 1), \mathbb{R})$ is closed in the topology of compact convergence which implies that the limit of f_n is continuous since each f_n is continuous. ■